Competent

38/40

95%

ELECTRICITY AND ELECTRONICS

FORMATIVE ASSESSMENT

Module #	671101000-KM-03:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

About the Formative Assessment	
Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ **points.** In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.

Special arrangements for assessment

Place	Eastc Electrical
Language	English
Resources	
Barriers	N/A

People to be involved with assessment

Learner: Sela Phiri	Manager:
Trainer: Miss R. Nyakala	Mentor / Coach:
Facilitator Miss R. Nyakala	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Sela Phiri

Learner's signature: S. Phiri Date:

Facilitator Name: Ruth Nyakala

Facilitator's signature: Date:

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

EASTC TECHNOCENTRIC VARSITY

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Sela Phiri	Signature S.
Facilitator Name: RUTH NYAKALA	Signature:

EASTC TECHNOCENTRIC VARSITY

LEARNING UNIT 1:

KM-03-KT01: Fundamentals of electricity (35%)

Learning Outcome 1:	Principles and fundamental concepts of electricity
Learning Outcome 2:	Definitions, types, properties and applications of conductors, insulators and semi-conductors
Learning Outcome 3:	Concepts, theories and principles of Electrical Circuits
Learning Outcome 4:	Calculations on basic electrical circuits (resistance, voltage and current) and power
Learning Outcome 5:	Basic principles and calculation of magnetism

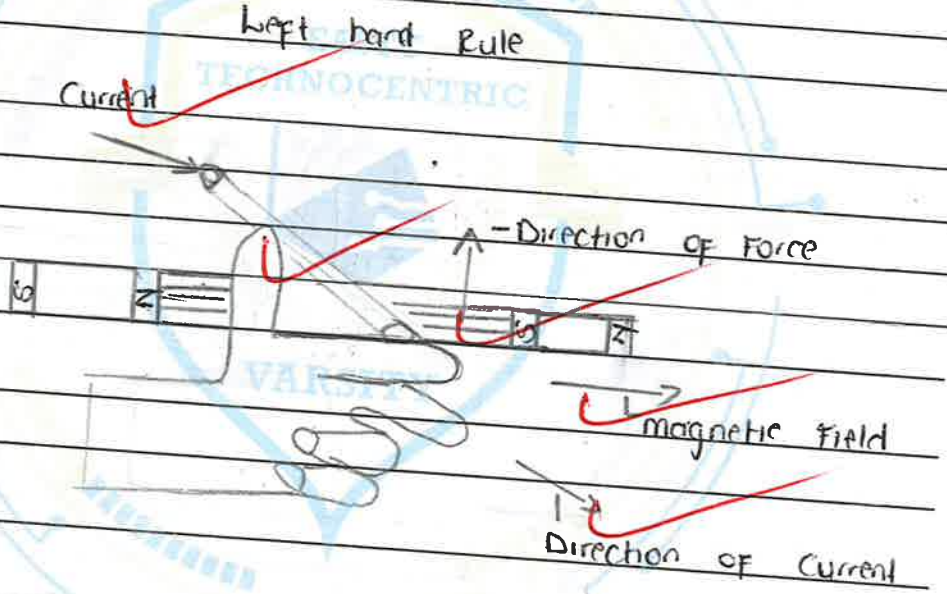
Task	Questions Description	Marks
1.	a. Describe, calculate and interpret fundamental concepts of electricity (Electro motive force, Potential difference, Resistance)using the correct units of measurement and definitions. b. Describe the factors that influences the resistance of a material c. Define and explain, using the correct units of measurement ohms law of electricity d. Define and explain using the correct units of measurement Kirchoff's law of electricity Manipulate formula to calculate voltage, current and resistance in series/parallel circuits	10

a.) Electromotive Force - is the total voltage generated by a source in an open circuit, representing the energy supplied per unit charge

Potential difference

* is the energy difference between two points in a circuit, driving current through a load

Task	Questions Description	Marks
2	a) Describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field. Fleming's left-hand rule must be demonstrated. b) Describe and explain the various forces or methods that can be used to alter magnetic fields, and describe the changes that take place	10



2.b. To alter a magnetic field, you essentially need to change the behavior of the sources - moving charges or intrinsic magnetic moments - that create the field.

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LEARNING UNIT 2:

KM-03-KT03: Earthing and bonding (35%)

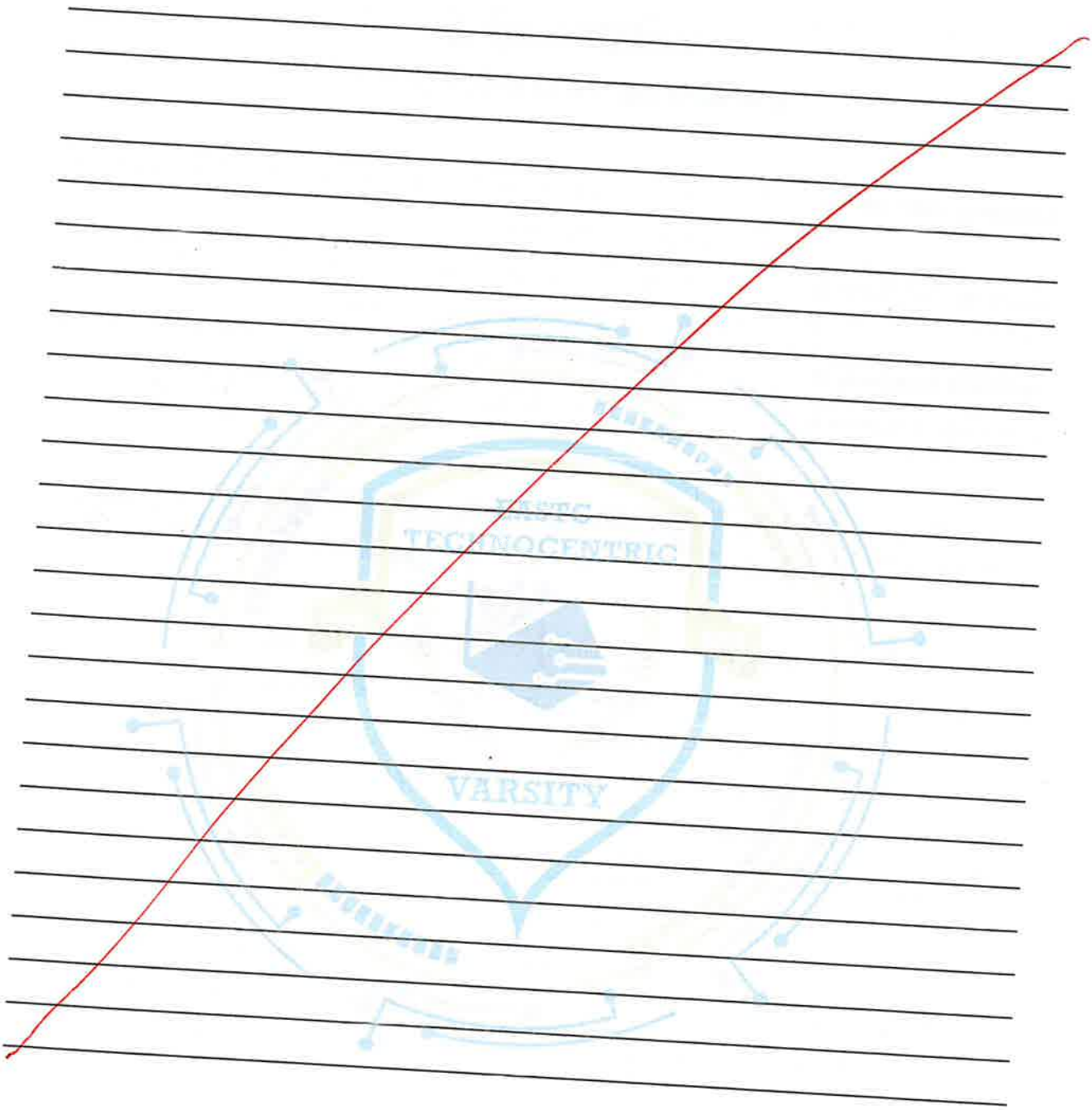
Learning Outcome 1:	KT0301 Principles of earthing and bonding
Learning Outcome 2:	KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.
Learning Outcome 3:	KT0303 Regulatory and statutory requirements related to earthing and bonding

Task	Questions Description	Marks
3	a) Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation b) Explain the term 'systems earthing' c) Describe the earthing systems relevant to SANS 10142-1. d) Describe how a common earth electrode is used in reticulation circuits	10

Regulatory requirements for earthing of neutral conductors
- The general rule in SANS 10142-1 is that the neutral conductor shall not be connected directly to earth or to the earth continuity conductor on the load side (consumer side) of the point of control
- Supplier side : The supply authority typically earths the neutral at the transformer to establish a reference point and maintain system stability

Consumer side : The consumer's neutral is generally kept isolated from earth within the installation downstream of the main switch

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Tantarum capacitors

-Stable, compact used in portable electronics

b Calculating capacitor value in parallel

$C = \text{Total} = C_1 + C_2 + C_3$

Calculating capacitor value in Series

$$C_{Total} = \frac{C_1 \times C_2}{C_1 + C_2}$$

Example Chart

Code	Calculation	Capacitance
101	10×10^1	100 pF
102	10×10^2	1000 pF = 1nF
103	10×10^3	10,000 pF = 10nF
104	10×10^4	100,000 pF = 0,1μF
105	10×10^5	1,000,000 pF = 1μF

C. Resistor Color Code Chart

Colour	number	multiplier	Tolerance
Black	0	$\times 10^0$	-
Brown	1	$\times 10^1$	± 1%
Red	2	$\times 10^2$	± 2%
orange	3	$\times 10^3$	-
Yellow	4	$\times 10^4$	-
Green	5	$\times 10^5$	± 0,5%
Blue	6	$\times 10^6$	± 0,25%
Violet	7	$\times 10^7$	±0,1%
~~white~~ Grey	8	$\times 10^8$	± 0,05%
White	9	$\times 10^9$	-
Gold	-	$\times 10^{-1}$	± 5%
Silver	-	$\times 10^{-2}$	± 10%

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ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS
Excellent

ASSESSMENT JUDGEMENT	
Learner's Total Mark: 95% **Requirements met** ☒ **not met** ☐	**Requirements**
Action/s required: N/A **By when:**	

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I ..Ruth Nyadala..... (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

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Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

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Assessment Planning

	Theory	**Practical**
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes
What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method. 1. Written Assignment : Printed Questionnaire 2. Observation : As per Practical Assessment Guide 3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation
What resources would you require to conduct the assessment? 1. Assessor toolkit 2. Theoretical assignment 3. Practical assessment 4. Learner toolkit
How would you sequence your activities? Explain 1. Record preparation process 2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 3. Learner will write summative assessments

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